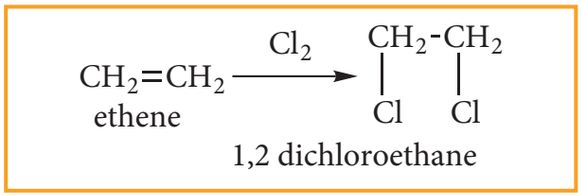
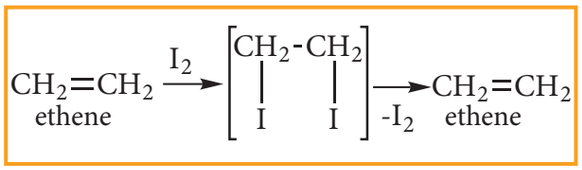
as catalytic hydrogenation. This process is of great importance in the manufacture of vanaspathi from vegetable oil. This helps to prevent rancidity of vegetable oils.

(ii) Addition of halogens: (Halogenation of alkenes)

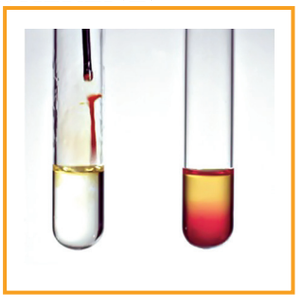
When alkene is treated with halogens like chlorine or bromine, addition takes place rapidly and forms 1, 2- dihalo alkane (or) vicinal dihalide.



Iodine reacts very slowly to form 1, 2 – diiodo alkane which are unstable and regenerate the original alkene by elimination of iodine.



TEST FOR ALKENE:



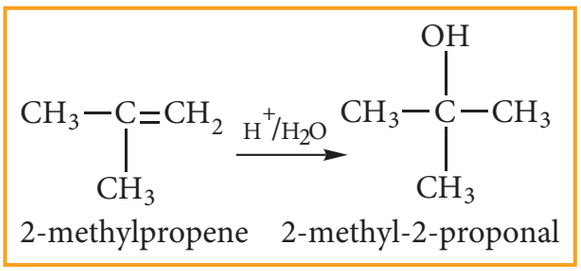
Bromine in water is reddish brown colour. When small amount of bromine water is added to an alkene, the solution is decolourised as it forms dibromo compound. So, this is the characteristic test for unsaturated compounds.

Markovnikoff's rule:

"When an unsymmetrical alkene reacts with hydrogen halide, the hydrogen adds to the carbon that has more number of hydrogen and halogen add to the carbon having fewer hydrogen". This rule can also be stated as in the addition reaction of alkene / alkyne, the most electro negative part of the reagent adds on to the least hydrogen attached doubly bonded carbon.

(iii) Addition of water:- (Hydration of alkenes)

Normally, water does not react with alkenes. In the presence of concentrated sulphuric acid, alkenes react with water to form alcohols. This reaction follows carbocation mechanism and Markovnikoff's rule.



(iv) Addition of hydrohalides: (Hydrohalogenation of Alkenes)

Hydrogen halides (HCl, HBr and HI) add to alkene to yield alkyl halides. The order of reactivity of different hydrogen halides is HI>HBr>HCl. It is an example for electrophilic addition.

(a) Addition of HBr to symmetrical alkene:

Addition of HBr to symmetrical alkene (similar groups are attached to double bond) yields alkyl halides (haloalkanes)

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